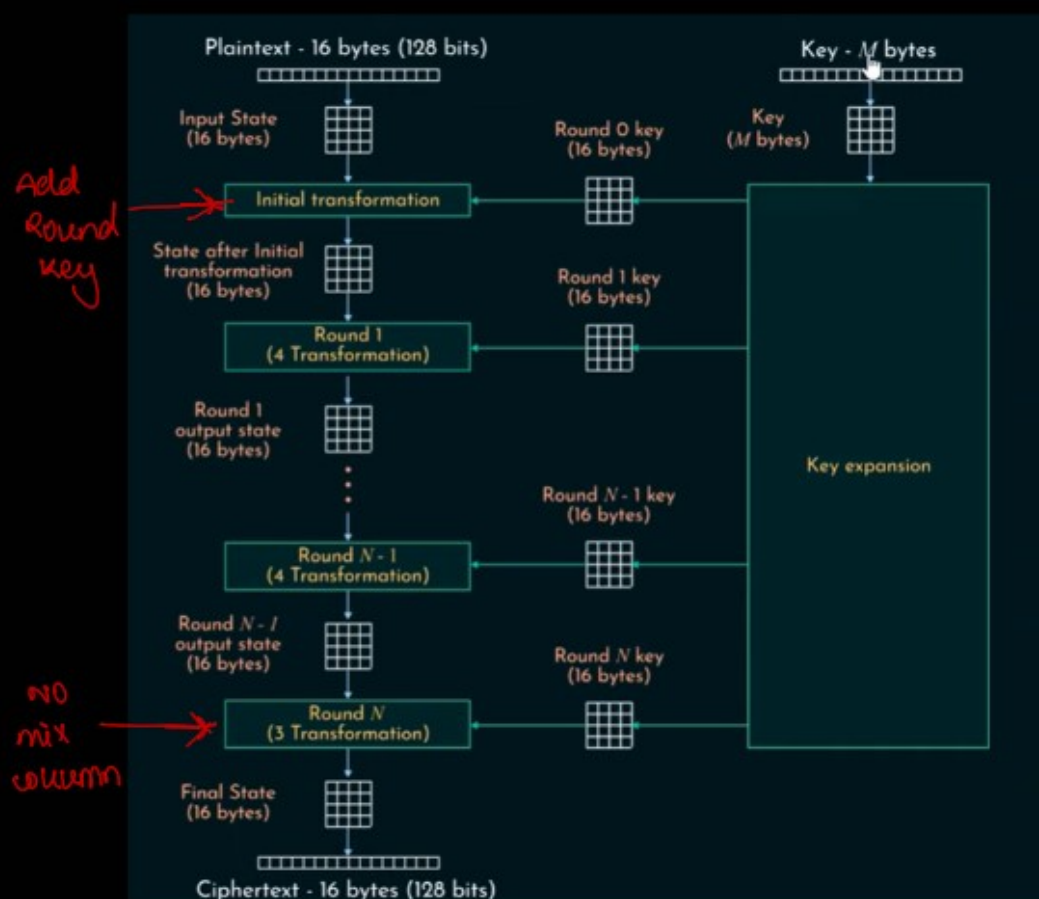


→ AES → Advanced encryption standard.
(128)



$M = 16B$
 $= 128 \text{ Bits}$
 $N = 10$
↓
For AES-128

↳ The 4 transformations are:

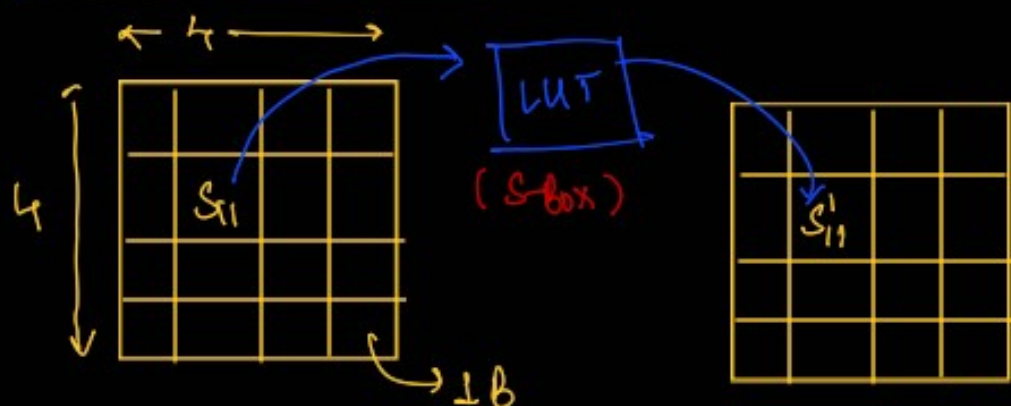
in this order →

- Sub-byte
- Shift rows
- Mix columns
- Add Round-Key

	AES-128
Key Size	128
Plaintext Size	128
Number of rounds	10
Round Key Size	128

→ AES functions :

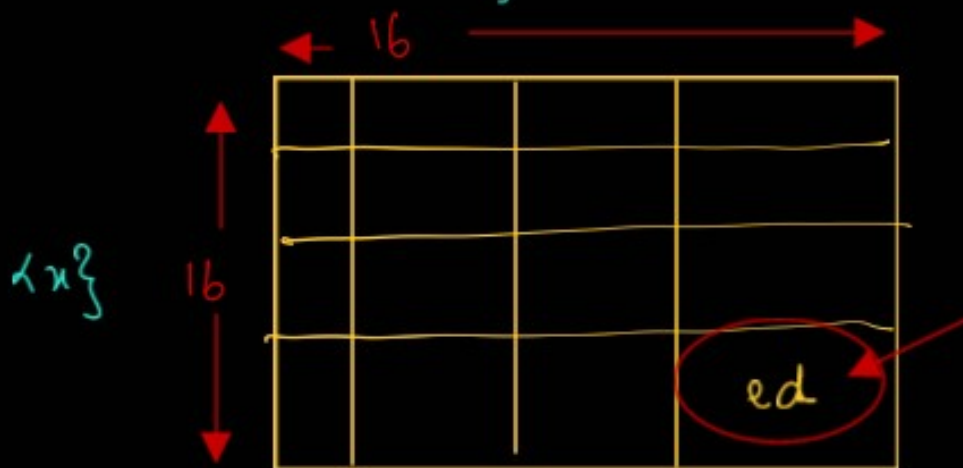
a) Substitute Bytes (substitution)



S-box : Fixed value defined by NIST. → Hardcoded memory

Fig : Each matrix → $4B \times 4B = 16B$
(2 hex digits)

$S_{11} = \{ \overset{x}{f} \overset{y}{f} \}$ → max possible val
 $\therefore x_{max} = f \mid y_{max} = f$



LUT value corresponding to $\{ff\}_{S_{11}}$

$\therefore S'_{11} = ed$

b) Shift Rows

s_{00}	s_{01}	s_{02}	s_{03}	→ left shift by 0
s_{10}	s_{11}	s_{12}	s_{13}	→ left shift by 1
s_{20}	s_{21}	s_{22}	s_{23}	→ left shift by 2
s_{30}	s_{31}	s_{32}	s_{33}	→ left shift by 3

} circular

c) Mix columns

$$\begin{bmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{bmatrix}$$

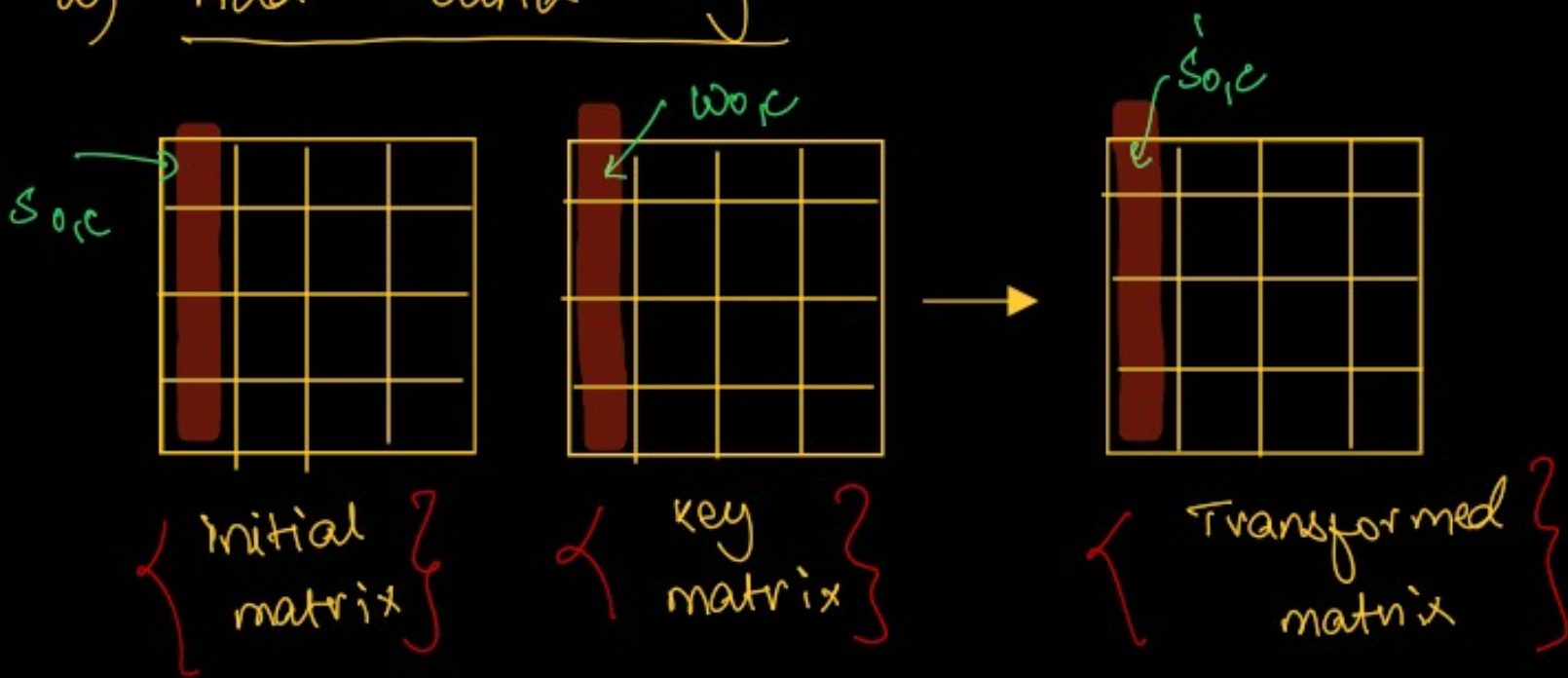
↓
new columnⁿ

↓
old columnⁿ

$$\therefore s_{0,c} = \{x02\} * s_{0,c} \oplus \{x03\} * s_{1,c} \oplus s_{2,c} \oplus s_{3,c}$$

** Addⁿ → XOR opⁿ

d) Add Round Key



$$s'_{0,c} = s_{0,c} \oplus w_{0,c}$$

→ Perform column wise XOR.

∴ For each round key f^n we need a key matrix which will have $4 \times 4 = 16B$
 $= 128 \text{ Bits}$
 $= 4 \text{ words}$

∴ Total key requirement =

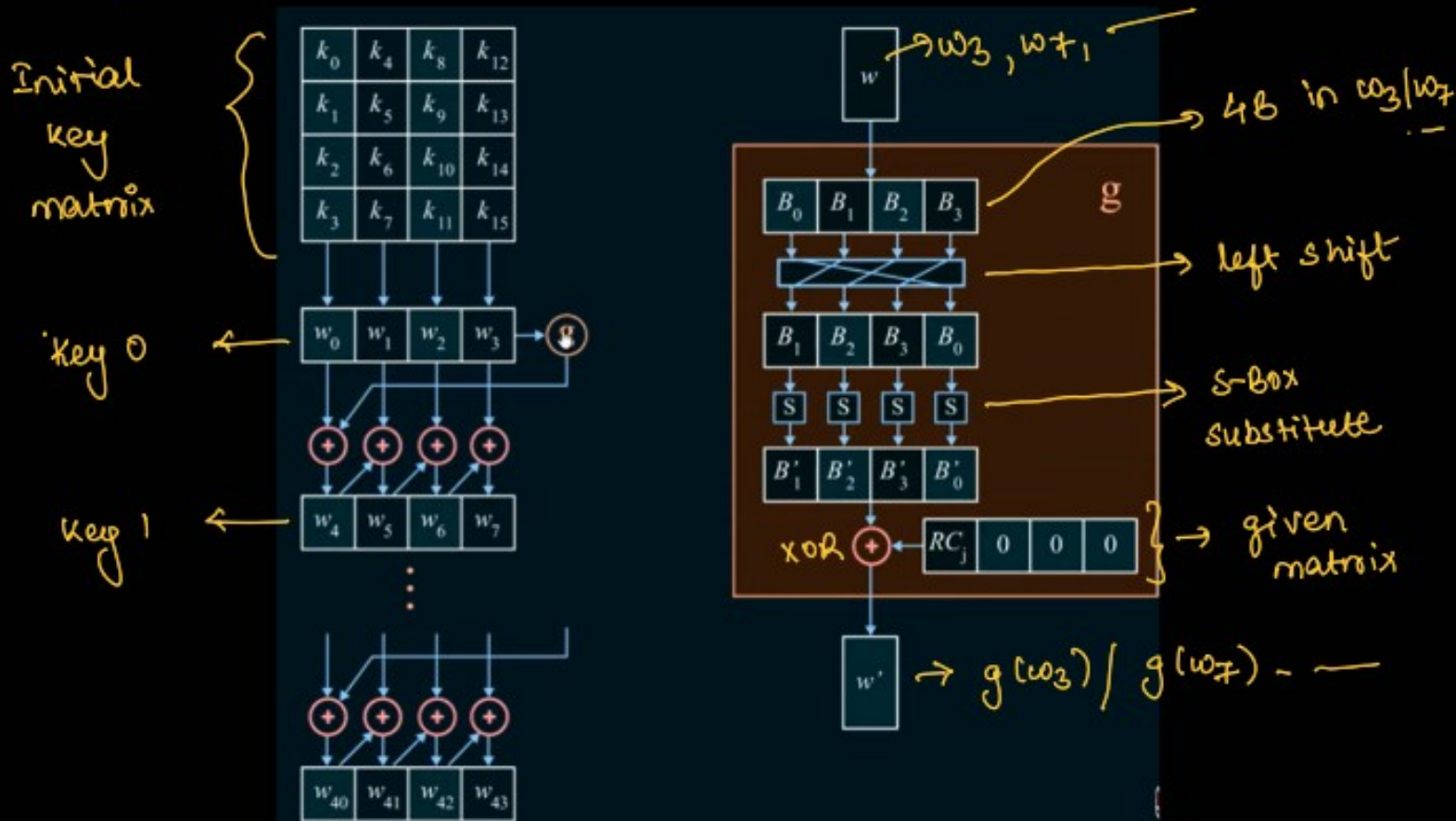
Initial transform + 10 Add Round Key

= (11×4) words

= 44 words [1 word = 32 Bits]

∴ 128 Bit is given as input

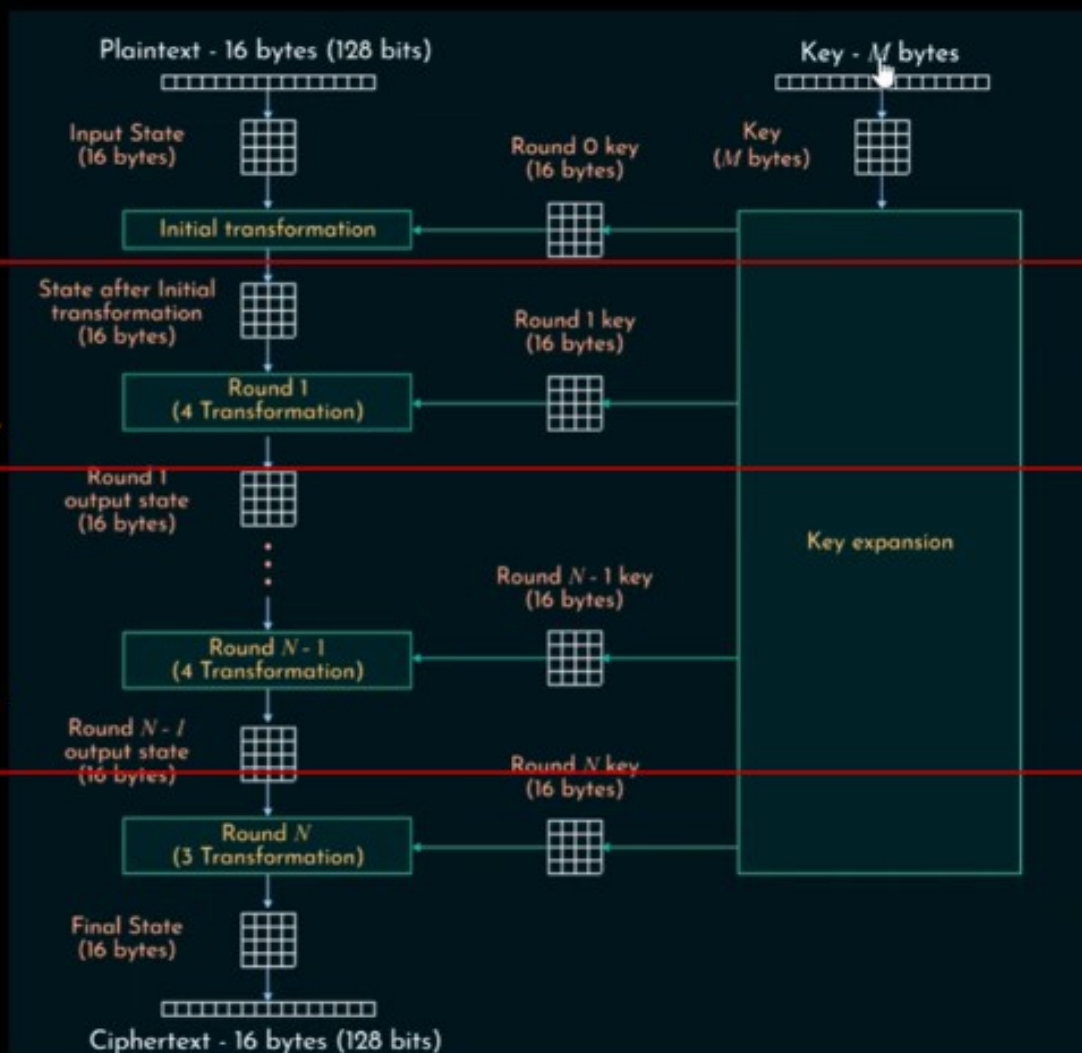
e) Key Expansion



→ simulation passed for simple combinational datapath.

→ Optimisation

- Scope to pipeline.



key is same for all the plaintext :lp.

∴ It has to be computed only once, no pipeline required for key exp.

↳ Just an initial overhead -

no. of stages in pipeline =

$$1 + 10 + 1 = 12 \text{ stag}$$

∴ to initial

for synchronous
op

→ going for 11-stage pipeline, not
 $1 + 9 \times 4 + 1 = 38$ stage
to balance area as well with
performance.

→ In our design, successfully implemented
12-stage pipeline

